Sample Questions

1. Let L_1 and L_2 be two languages decided by Turing machines M_1 and M_2, respectively. Design a Turing machine M using M_1 and M_2 that decides $L_1 \cap L_2$ or justify if it is not possible.

2. Let $L = \{\langle G, s, t\rangle \mid G$ is a directed graph such that there is a path from s to t $\}$. Prove that L is in class P. Prove that L is in class NP.

3. Let $T = \{\langle G\rangle \mid G$ is a tree$\}$. Prove that T is in class P. Prove that T is in class NP.

4. If L is a language over some alphabet Σ^* then the complement of L denoted as $\overline{L}$ is defined as $\overline{L} = \Sigma^* - L$. We say a language L is in the class coNP if its complement $\overline{L}$ is the class NP. Prove that $\mathsf{NP} = \mathsf{coNP}$ if and only if $A \leq_P B$ and $B \leq_P A$ for two NP-complete languages A and B.

5. Show that $2\textsc{sat}$ is in class P.

6. Show that there exists a Turing-unrecognizable language.

7. Prove/disprove that if a language L is Turing-decidable then $\overline{L}$ is not necessarily a Turing-decidable language.

8. Show that the class NP is closed under the star operation. That if L is in NP then L^* is also in NP.

9. Let $T = \{\langle M\rangle\ M$ is a TM that accepts w^{rev} whenever it accepts $w\}$. Show that T is undecidable.

10. If $A \leq_P B$ and $B \leq_P C$ then $A \leq_P C$. So if A is NP-complete and C is also NP-complete then B must also be NP-complete.